ORIENTATION TO COMPUTING

UNIT 6: CLOUD COMPUTING, VIRTUALIZATION & CYBERSECURITY ESSENTIALS

Curriculum: Lovely Professional University | School of Computer Science & Engineering
Reference: Cloud Computing: Principles and Paradigms by Rajkumar Buyya & Cryptography and Network Security by William Stallings

>> 6.1 Fundamentals and Architectural Models of Cloud Computing

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned with minimal management effort:
* 5 Essential Characteristics (NIST Special Publication 800-145):
- 1. On-Demand Self-Service: Consumers provision server time and network storage automatically without human interaction with service providers.
- 2. Broad Network Access: Capabilities accessible over network through standard heterogeneous client platforms (laptops, mobile phones, thin clients).
- 3. Resource Pooling: Provider computing resources pooled to serve multiple consumers using a multi-tenant model with dynamic physical/virtual reassignment.
- 4. Rapid Elasticity: Capabilities elastically provisioned and released to scale outward and inward commensurate with demand (auto-scaling).
- 5. Measured Service: Resource usage monitored, controlled, and reported, providing transparency and Pay-As-You-Go billing.
* The 3 Primary Cloud Service Models (SPI Tier):
- IaaS (Infrastructure as a Service): Provides raw compute VMs, storage, networks. Customer manages OS, runtime, and apps (AWS EC2, Google Compute Engine, Azure VMs).
- PaaS (Platform as a Service): Provides hardware, OS, and development runtime frameworks. Customer deploys application code without managing underlying servers (Google App Engine, Heroku, AWS Elastic Beanstalk).
- SaaS (Software as a Service): Complete cloud application accessible via web browser. Customer manages only user data and settings (Google Workspace, Microsoft 365, Salesforce).
* Cloud Deployment Models: Public Cloud (AWS, GCP), Private Cloud (dedicated internal infrastructure), Hybrid Cloud (combining public and private via secure VPN/DirectConnect), Community Cloud.

>> 6.2 Virtualization Technologies: Hypervisors vs Containers

* Virtualization: The abstraction of physical computer hardware via software, enabling multiple isolated operating system environments to execute concurrently on a single physical host:
* Type-1 Hypervisor (Bare-Metal): Runs directly on physical host hardware without an underlying host OS. Maximum performance, lowest latency, enterprise standard (VMware ESXi, Microsoft Hyper-V, KVM, Xen).
* Type-2 Hypervisor (Hosted): Runs as a software application on top of a standard host OS. Higher overhead (VirtualBox, VMware Workstation).
* Containerization (OS-Level Virtualization - Docker):
- Core Paradigm: Containers share the host operating system kernel directly, packaging only user-space binaries, libraries, and application dependencies.
- Linux Kernel Primitives: Namespaces (providing isolated views of PID, Network, Mount points, IPC) and Cgroups (enforcing resource limits on CPU, memory, I/O bandwidth).
- Advantages over VMs: Lightweight (megabytes vs gigabytes), near-instantaneous startup (milliseconds vs minutes), high server density, unified dev-to-prod parity.

>> 6.3 Cybersecurity Fundamentals: The CIA Triad and Cryptography

* The CIA Triad (The Core Tenets of Information Security):
- Confidentiality: Ensuring sensitive data is accessible strictly to authorized entities (enforced via Encryption, Access Control Lists).
- Integrity: Ensuring information is accurate and has not been maliciously or accidentally tampered with (enforced via Cryptographic Hash Functions: SHA-256, HMAC).
- Availability: Ensuring computing services and data remain accessible to authorized users when needed (enforced via Redundancy, DDoS Mitigation, Fault Tolerance).
* Cryptographic Paradigms:
- Symmetric Key Cryptography: Same secret key used for encryption and decryption. Fast, high throughput (AES-256, ChaCha20). Key exchange problem is major challenge.
- Asymmetric Key (Public Key) Cryptography: Mathematically linked key pair. Public key encrypts data; matching Private key decrypts data (RSA, Elliptic Curve Cryptography - ECC). Used for key exchange, digital signatures, SSL/TLS certificates.
- Cryptographic Hash Functions: One-way deterministic mathematical functions producing fixed-size digest (e.g., SHA-256). Pre-image resistant, second pre-image resistant, collision resistant.

>> 6.4 Common Cyber Threats, Attack Vectors and Defenses

* Malware Taxonomies: Viruses (attach to host files), Worms (self-replicating across networks without host intervention), Trojan Horses (disguised malicious utilities),

Ransomware (encrypts user files and demands cryptocurrency ransom), Spyware/Keyloggers.
* Network Attacks: Man-in-the-Middle (MitM - eavesdropping and injecting packets), Denial of Service / Distributed Denial of Service (DoS/DDoS - flooding targets with botnet traffic), Phishing / Social Engineering.
* Defense Mechanisms: Stateful Inspection Firewalls, Intrusion Detection & Prevention Systems (IDS/IPS), Multi-Factor Authentication (MFA), Zero Trust Architecture (Never Trust, Always Verify).

>> 6.5 Emerging Computing Paradigms: IoT, AI Infrastructure & Blockchain

* Internet of Things (IoT): Constellation of physical devices embedded with sensors, processing power, and connectivity to exchange telemetry via lightweight protocols (MQTT, CoAP).
* Edge Computing: Processing data locally near the point of generation (at the network edge/gateway) rather than routing all raw streams to centralized cloud datacenters, minimizing latency and bandwidth consumption.
* Blockchain & Distributed Ledgers: Immutable append-only chain of cryptographically linked blocks maintaining consensus across decentralized peer-to-peer networks (Proof of Work, Proof of Stake).

[PROBLEM 1: Containerization vs Virtual Machines Performance Benchmark]

Problem Statement: An engineering team compares deploying 100 microservice instances using Virtual Machines (VMs running on a Type-1 hypervisor) versus Docker containers on Linux. Quantitatively compare: (1) Storage footprint, (2) Memory overhead, and (3) Startup latency.
Architectural Performance Matrix:
* 1. Storage Footprint Comparison:
- VM Approach: Each VM requires a complete Guest OS image (Ubuntu ~2.5 GB minimum) plus application (~50 MB). Total storage = 100 x 2.55 GB = 255 GB.
- Container Approach: Base Alpine/Debian image shared via union filesystem; application image layer = ~60 MB. Total storage = ~2.5 GB base + (100 x 60 MB delta) = ~8.5 GB (a 30x storage reduction!).
* 2. RAM Memory Consumption:
- VM: Each Guest OS kernel and system daemons require minimum 512 MB reserved RAM before application starts. Total idle RAM = 100 x 512 MB = 51.2 GB.
- Container: Shares single host kernel. Containers consume zero idle kernel overhead; only active application memory (~25 MB each). Total RAM = 100 x 25 MB = 2.5 GB (a 20x

RAM savings!).
* 3. Startup Latency:
- VM: Cold OS boot, BIOS/UEFI emulation, systemd daemon initialization = 30 to 60 seconds.
- Container: Creating kernel namespaces and launching process = 50 to 200 milliseconds (300x faster).

[PROBLEM 2: RSA Public Key Cryptosystem Mathematical Computation]

Problem Statement: Demonstrate RSA Public Key cryptography end-to-end using prime numbers p = 11 and q = 13. (1) Calculate modulus n and Euler totient phi(n), (2) Choose public key exponent e = 7 and calculate private key exponent d, and (3) Encrypt plaintext message M = 9, then decrypt the resulting ciphertext C.
Step-by-Step Mathematical Derivation:
* 1. Calculate Modulus and Totient:
- Modulus: n = p x q = 11 x 13 = 143.
- Euler Totient: phi(n) = (p - 1) x (q - 1) = (11 - 1) x (13 - 1) = 10 x 12 = 120.
* 2. Key Generation:
- Public Exponent: e = 7 (Check: gcd(7, 120) = 1, valid coprimes). Public Key = (e, n) = (7, 143).
- Private Exponent (d): Modular multiplicative inverse: (d x e) mod phi(n) = 1 -> (d x 7) mod 120 = 1.
- Solving via Extended Euclidean Algorithm: 120 = 7 x 17 + 1 -> 1 = 120 - 7 x 17 = 120 x (1) + 7 x (-17).
- d = -17 mod 120 = 120 - 17 = 103. Check: (103 x 7) = 721 = (6 x 120) + 1 = 1 mod 120 (Verified!).
- Private Key = (d, n) = (103, 143).
* 3. Encryption of Message M = 9:
- Ciphertext C = (M^e) mod n = (9^7) mod 143.
- 9^2 = 81 mod 143 | 9^4 = 81^2 = 6561 = 45 x 143 + 126 = 126 mod 143 = -17 mod 143.
- 9^7 = 9^4 x 9^2 x 9 = (-17) x 81 x 9 = (-17) x 729 = (-17) x (5 x 143 + 14) = -17 x 14 = -238.
- -238 mod 143 = -238 + 2 x 143 = -238 + 286 = 48.
- Ciphertext C = 48.
* 4. Decryption of Ciphertext C = 48:
- Plaintext M = (C^d) mod n = (48^103) mod 143 = 9 (Original message recovered perfectly!).

[PROBLEM 3: Defense-in-Depth Cybersecurity Architecture Design]

Problem Context: Design an enterprise multi-tier cybersecurity architecture to defend an electronic health records (EHR) database against modern Ransomware attacks.

Tiered Defense Implementation Protocol:

* 1. Perimeter Tier (Network Ingress): Next-Generation Firewall (NGFW) with deep packet inspection; Web Application Firewall (WAF) blocking SQL injection and cross-site scripting (XSS); DDoS mitigation shield.
* 2. Network Demilitarized Zone (DMZ): Isolate public-facing web servers from backend databases. Zero-trust network microsegmentation preventing lateral lateral movement between subnets.
* 3. Endpoint Tier: Endpoint Detection and Response (EDR) software with behavioral heuristic monitoring to detect unauthorized bulk cryptographic file modifications; mandatory application whitelisting.
* 4. Identity & Access Management (IAM): Role-Based Access Control (RBAC); Multi-Factor Authentication (MFA) via FIDO2 hardware keys; strictly adhere to the Principle of Least Privilege (PoLP).
* 5. Data & Storage Tier: End-to-end AES-256 encryption for data at rest and TLS 1.3 for data in transit; immutable, air-gapped write-once-read-many (WORM) offline backups to guarantee recovery without paying ransom.

[PROBLEM 4: Edge Computing vs Cloud Latency Analysis in Autonomous Systems]

Problem Statement: An autonomous vehicle traveling at 100 km/h (27.78 m/s) utilizes high-resolution LiDAR and cameras to detect unexpected road obstacles. Compare obstacle reaction distance between: (1) Local On-Board Edge Processing (latency 15 ms), and (2) Cloud Datacenter Processing (uplink + compute + downlink latency 180 ms).

Mathematical Derivation:

* 1. Reaction Distance Formula: Distance = Velocity (m/s) x Total Decision Latency (s).
* 2. Local Edge Computing Processing:
- Total latency = 15 ms = 0.015 seconds.
- Reaction distance = 27.78 m/s x 0.015 s = 0.417 meters (approx 42 cm). Vehicle initiates emergency braking within less than half a meter of obstacle detection.
* 3. Centralized Cloud Datacenter Processing:
- Total latency = 180 ms = 0.180 seconds.
- Reaction distance = 27.78 m/s x 0.180 s = 5.00 meters.
* 4. Engineering Impact: Centralized cloud processing introduces an unacceptable 5-meter blind delay before brakes can be applied, proving why life-critical autonomous systems mandate localized edge computing.